

Kenuo Xu

SENIOR ENGINEER · HUAWEI

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Employment

Huawei

Shanghai, China

SENIOR ENGINEER

2025.08 - Present

- Working in the WLAN Lab (Data Communication Product Line), focusing on AI for CSI-based sensing and physical-world understanding (received an “Excellent” rating upon completion of probation).
- Spearheaded the design and development of end-to-end AI algorithm architectures, leading the pre-research and deployment of intelligent sensing models for smart security and smart building scenarios.
- Developed lightweight CNN-based sensing models for practical deployments, incorporating anomaly detection, data augmentation, and ADDA-based domain adaptation to address limited labels, data imbalance, and environmental variations.
- Designed a self-supervised Wi-Fi CSI pretraining framework with domain-specific tokenization and physics-guided objectives, enabling transferable representations for fine-tuning across sensing environments and downstream tasks.
- Worked cross-functionally to build closed-loop data pipelines spanning collection, cleaning, augmentation, and evaluation, while optimizing and deploying sensing models on edge devices for real-world validation and product integration.

Microsoft Research Asia

Shanghai, China

RESEARCH INTERN

Dec. 2022 - Sep. 2023

- In the Shanghai Wireless Group; mentor: Prof. Lili Qiu
- Conducted cross-disciplinary research at the intersection of Large Language Models (LLMs) and computer networking systems, exploring the potential of applying LLMs to underlying system optimizations.
- Designed and implemented an LLM-driven intelligent parameter-tuning framework for network algorithms, leveraging the model’s comprehension and reasoning capabilities to replace manual tuning.

Education

Peking University

Beijing, China

PH.D. IN COMPUTER SCIENCE

Sep. 2020 - Jun. 2025

- In the Software-hardware Orchestrated ARchitecture (SOAR) group; advisor: Prof. Chenren Xu
- Research interest: algorithms of mobile computing and their applications in IoT, robotics, and human-computer interaction.
- Designed a visible light backscatter communication system that supports concurrent transmission for low latency purpose.
- Designed a visible light communication system with spike cameras as receivers to achieve high data rate and dynamic range.
- Designed a liquid-crystal fiducial marker system using LiDAR as receivers for extended reading range and higher ranging accuracy.
- See publications for more research.

Peking University

Beijing, China

B.SC. IN COMPUTER SCIENCE

Sep. 2016 - Jun. 2020

- Graduated with Excellent Graduate Award.

Publications

RetroLiDAR: A Liquid-crystal Fiducial Marker System for High-fidelity Perception of Embodied AI

ACM SenSys

KENUO XU, BO LIANG, JINGYU LI, CHENREN XU

2025

- A long-range high-ranging-accuracy fiducial marker system using LiDAR and liquid crystals for robotics and virtual reality.

RetroV2X: A New V2X Paradigm with Visible Light Backscatter Networking

Fundamental Research

CHENREN XU, KENUO XU, LILEI FENG, BO LIANG

2023

- A practical vehicle-to-everything communication system with visible light.

When Visible Light (Backscatter) Communication Meets Neuromorphic Cameras in V2X

ACM HotMobile

KENUO XU, KEXING ZHOU, CHENGXUAN ZHU, SHANGHANG ZHANG, BOXIN SHI, XIAOQIANG LI, TIEJUN HUANG, CHENREN XU

2023

- When VLC meets neuromorphic cameras: a spike cameras as VLC receiver achieves 4.8 kbps data rate and different mobile scenarios.

Low-Latency Visible Light Backscatter Networking with RetroMUMIMO

ACM SenSys

KENUO XU, CHEN GONG, BO LIANG, YUE WU, BOYA DI, LINGYANG SONG, CHENREN XU

2022

- Enables 8 concurrent VLBC links and reduces networking latency by 92.0%.

